A Synthetic Chemoinformatics Testbed for Evaluating Model Interpretability and Green Catalyst Prioritization

A Preliminary Computational Technical Note

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Abstract

As the chemical sciences rapidly adopt descriptor-based machine learning, transparent validation environments remain critical. This technical note introduces a proof-of-concept synthetic testbed of 5,000 catalytic alcohol oxidation reactions. By mathematically defining the data-generating process via an additive heuristic, this testbed allows for the explicit auditing of machine learning pipelines. Evaluated across 20 independently seeded synthetic datasets, we benchmark Linear Regression against tree-based ensembles and Support Vector Regression (SVR) relative to the empirical oracle noise floor. Linear regression matches the empirical oracle error floor, while the other models exhibit higher errors. A controlled ablation conducted specifically on the stated, untuned RBF-SVR configuration demonstrated that including non-causal continuous decoy variables degraded its predictive accuracy. Furthermore, we utilize TreeSHAP to show that the fitted XGBoost model assigned lower global attribution to the decoy features on the reference dataset. Finally, we implement an author-defined Green Performance Index (GPI) as an illustration of a post-hoc decision rule that penalizes higher reaction temperatures.

1 Introduction and Methodology

To bypass the computational bottlenecks of Density Functional Theory (DFT), modern approaches utilize descriptor-based predictive modeling [1, 2]. However, highly parameterized models can sometimes maintain apparent performance while relying on non-causal or spurious representations [2, 3]. Controlled ground-truth benchmarks are therefore useful as a methodological sanity check for model evaluation and feature attribution [4].

1.1 Data Generating Process (DGP) and Sampling Probabilities

Datasets of $N = 5,000$ synthetic reactions were generated across 20 independent random seeds (Seeds 42 through 61). For each reaction, categorical identities were sampled independently from discrete uniform distributions: substrate identity Substrate_Idx $\sim$ Uniform$\{0, \ldots, 21\}$ ($p = 1/22$ per substrate), catalyst identity Cat_Idx $\sim$ Uniform$\{0, 1, 2, 3\}$ ($p = 1/4$ per catalyst), and oxidant identity Ox_Idx $\sim$ Uniform$\{0, 1, 2\}$ ($p = 1/3$ per oxidant). Reaction temperature was sampled continuously via $T \sim \mathcal{U}(25, 110)$°C. The target yield (in yield percentage points) was defined by the additive heuristic function:

$$Y_{\text{raw}} = \text{Base_Cat} + \text{Base_Ox} - 0.05M + 0.15(T - 25^{\circ}\text{C}) + \varepsilon \tag{1}$$

$$Y = \min(100, \max(0, Y_{\text{raw}})) \tag{2}$$

Where Base_Cat and Base_Ox represent categorical baseline efficiencies, M is a molecular-mass term derived from the RDKit exact monoisotopic mass (`ExactMolWt`, in Da), T represents thermal

activation above 25°C, and $\varepsilon \sim \mathcal{N}(0, 5^2)$ introduces Gaussian noise with a standard deviation of $\sigma = 5$ percentage points.

For unbounded Gaussian noise, the analytical expected absolute error is $E[|\varepsilon|] = \sigma\sqrt{2/\pi} \approx 3.989\%$. To account for boundary clamping in Equation 2, we also compute the empirical oracle predictor $\hat{Y}_{\text{oracle}} = \min(100, \max(0, \text{Base_Cat} + \text{Base_Ox} - 0.05M + 0.15(T - 25°\text{C})))$. Because an average of only 2.80 out of 5,000 reactions per dataset exceed $[0, 100]\%$ before clamping, the 20-seed empirical oracle MAE is $3.994 \pm 0.043\%$ ($R^2 = 0.870 \pm 0.003$; on the Seed 42 reference dataset, empirical oracle MAE is 4.060%, $R^2 = 0.866$). To test sensitivity to uninformative inputs, the feature matrix included three independent uniform noise variables: `Decoy_TPSA`, `Decoy_Rot_Bonds`, and `Decoy_Cat_Load`.

2 Results and Discussion

2.1 Multi-Seed Baseline Recovery and SVR Decoy Ablation

Models were evaluated using fixed hyperparameter configurations across a 5-fold cross-validation protocol under both Random KFold and substrate-grouped hold-out (GroupKFold). Within each fold, continuous features were standardized using a `StandardScaler` fitted exclusively on that fold's training split and applied to the held-out test split. For each dataset seed, the five fold-level MAE and R^2 scores were averaged first; Tables 1 and 2 report the mean and sample standard deviation across the 20 seed-level averages.

Table 1: Model performance relative to oracle references (Mean ± Sample Standard Deviation of fold-averaged scores across 20 independent datasets, Seeds 42–61).

Model / Reference	**Validation Strategy**	**MAE (%)**	**R^2**
Theoretical Unbounded Gaussian Reference	Analytical ($5\sqrt{2/\pi}$)	3.989	N/A
Empirical Oracle (Clipped Noiseless Mean)	Full-Dataset Reference	3.994 ± 0.043	0.870 ± 0.003
Linear Regression	Random CV	4.001 ± 0.044	0.869 ± 0.003
XGBoost	Random CV	4.106 ± 0.043	0.862 ± 0.003
Random Forest	Random CV	4.496 ± 0.040	0.834 ± 0.003
SVR (Untuned, With Decoys)	Random CV	4.488 ± 0.038	0.834 ± 0.003
Linear Regression	GroupKFold (Substrate Hold-Out)	4.001 ± 0.046	0.869 ± 0.003
XGBoost	GroupKFold (Substrate Hold-Out)	4.132 ± 0.044	0.860 ± 0.003
Random Forest	GroupKFold (Substrate Hold-Out)	4.538 ± 0.038	0.831 ± 0.003
SVR (Untuned, With Decoys)	GroupKFold (Substrate Hold-Out)	4.553 ± 0.055	0.828 ± 0.005

Table 2: Paired SVR Decoy Ablation Study (Random CV across 20 datasets, fixed $C = 1.0, \epsilon = 0.1$)

Model Configuration	**MAE (%)**	**R^2**
SVR (Untuned, With 3 Decoys)	4.488 ± 0.038	0.834 ± 0.003
SVR (Untuned, Without Decoys)	4.128 ± 0.044	0.860 ± 0.003

Linear regression closely approaches the empirical oracle noise floor (4.001 ± 0.044% vs. 3.994 ± 0.043%), while the other models exhibit higher errors. Performance remained comparable under substrate-grouped validation, as expected when substrate identity enters the additive DGP solely through `ExactMolWt`. In the paired ablation conducted for SVR (Table 2), removing the three synthetic decoys reduced MAE from 4.488 ± 0.038% to 4.128 ± 0.044%. This sensitivity applies specifically to the stated, untuned RBF-SVR configuration ($C = 1.0, \epsilon = 0.1$), where isotropic

Euclidean distances are diluted by standardized uninformative continuous features; paired decoy ablations were not conducted for the tree-based models.

2.2 Isolating Signal from Decoys via TreeSHAP

SHAP values were computed using `shap.TreeExplainer` on an XGBoost model fitted to the full reference dataset (Seed 42), summing absolute SHAP values across one-hot encoded columns for categorical features.

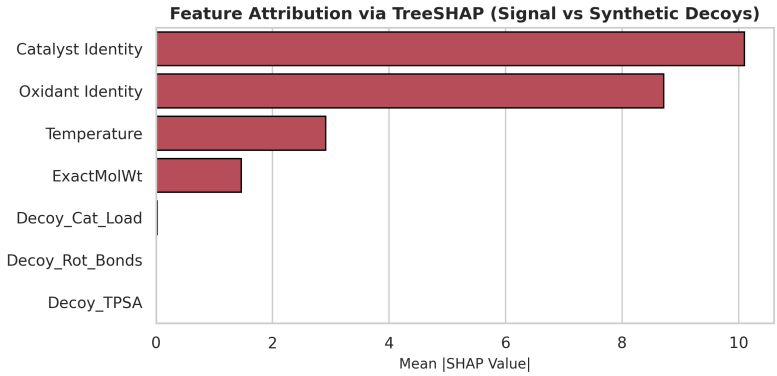


Figure 1: TreeSHAP Feature Attribution on the Seed 42 reference dataset comparing target-generating inputs against synthetic decoys.

The fitted XGBoost model assigned lower global attribution to the decoy features (`Decoy_Cat_Load`, `Decoy_Rot_Bonds`, `Decoy_TPSA`) on the Seed 42 reference dataset. While this illustrates separation between target-generating and decoy inputs in the fitted model, it does not establish causal correctness or out-of-sample attribution stability across seeds.

2.3 Illustration of a Decision Rule: Green Performance Index (GPI)

As an illustration of a post-hoc multi-criteria decision rule on the **Seed 42 reference dataset**, an author-defined GPI was computed using the **observed, noisy, clamped yield (Y, `Target_Yield`)**, catalyst selectivity (with additive Gaussian noise $\varepsilon_{\text{sel}} \sim \mathcal{N}(0, 2^2)$ in selectivity percentage points), and oxidant sustainability. The predominance of low-temperature reactions in Table 3 is a direct mathematical consequence of the chosen temperature penalty (-0.5 sustainability points per °C outweighing the $+0.15$ yield boost per °C). The GPI is an illustrative heuristic, not an experimentally validated measure of environmental impact.

Table 3: Top 10 Synthetic Configurations Ranked by GPI on the Seed 42 Reference Dataset (`synthetic_benchmark_dataset.csv`), computed using observed noisy yields (Y).

Reaction ID	Catalyst	Oxidant	Temp (°C)	Observed Yield (%)	Norm GPI
RXN_1601	Mo-Het	O2	25.6	90.8	4.642
RXN_1598	Mo-Het	O2	31.4	89.9	4.580
RXN_3035	Mo-Het	O2	36.1	91.4	4.579
RXN_4503	Mo-Het	O2	25.2	87.7	4.565
RXN_1468	Mo-Het	O2	41.3	91.1	4.546
RXN_1463	Mo-Het	O2	28.0	82.9	4.533
RXN_3024	Mo-Het	O2	44.3	91.1	4.524
RXN_3446	Mo-Het	O2	32.0	85.2	4.523
RXN_3673	Mo-Het	O2	26.1	85.2	4.523
RXN_1615	Mo-Het	O2	29.0	85.0	4.517

3 Conclusion, Limitations, and Declarations

This preliminary technical note provides a reproducible synthetic testbed for auditing chemoinformatics machine learning workflows. In this additive setting, linear regression recovers the empirical oracle noise floor, and untuned RBF-SVR degrades when exposed to uninformative continuous decoys.

Scope and Limitations: This benchmark is a synthetic proof-of-concept built on 22 substrate structures, an additive target equation without substrate-specific random effects, and independent uniform decoy variables. It does not establish predictive accuracy for experimental alcohol-oxidation chemistry.

Data and Code Availability: The Python generation and 20-seed evaluation scripts, the Seed 42 reference dataset (`synthetic_benchmark_dataset.csv`), and the 20-seed metric log (`20_seed_metrics_log.csv`) are archived on Zenodo (concept DOI: 10.5281/zenodo.23383768) and GitHub (project repository). All reactions and labels are synthetic; results are traceable and consistency-checked, not independently reproduced in a clean environment.

References

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